Machine Learning Applications in Financial Leasing Default Risk Prediction

Group Id - 25-26J-179

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B.Sc. (Hons) Degree in Information Technology Specialization in Information Technology

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DECLARATION

I declare that this is my own work, and this dissertation does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text. Also, I hereby grant to Sri Lanka Institute of Information Technology the non-exclusive right to produce and distribute my dissertation in whole or part in print, electronic or other medium. I retain the right to use this content in whole or part in future works (such as article or books).

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ABSTRACT

This research focuses on a machine-learning driven credit default prediction system that is employed as an essential tool in the financial leasing industry to assist with identifying potential non-payment customers through the use of their financial and behavioral data so that financial organizations can make more precise and timely decisions regarding their credit's risk assessment.

The default risk prediction model used a structured data processing and prediction pipeline for the purposes of classifying customers based on their likelihood of failing to meet their payment obligations, which was accomplished through the use of procedures such as data cleansing, feature engineering[9], as well as standard scaling of the data, which consisted of raw leasing transaction data including the amount financed, loan term, interest rate, amount of rent and amount of any overdue payments made by that customer, as well as customer's demographic information. In order to create an ensemble model that would allow for improving the overall predictive accuracy over the modelling methodologies of Random Forest [1], CatBoost and XGBoost [2], the individual component models produced accuracies of 0.9953, 0.9955 and 0.9954 respectively, however, the stacking ensemble model[5] produced the highest prediction accuracy of 0.9956.

The predicted Probability of Default (PD) provides clientele with three levels of risk: High Risk (PD ≥ .80), Medium Risk (.20 ≤ PD < .80), or Low Risk (PD <.20). These classifications facilitate proactive decision making and help to improve risk management procedures for financial leasing businesses.

Additionally, this system has been developed as a web-based application using FastAPI, allowing for real-time predictions and ease of integration. Through the use of Pydantic, data is validated; while through the use of Pandas and NumPy, data is processed in an efficient manner. The proposed solution provides a reproducible, scalable method for predicting default risk, enabling leasing companies to reduce their exposure to financial loss and increase their operational effectiveness.

Keywords: Default Risk Prediction, Financial Leasing, Machine Learning, Stacking Ensemble[5]***, Credit Risk*** [3]

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1 INTRODUCTION

1.1 Background Literature

The finance lease business is a credit-dependent business where the accurate evaluation of a customer's ability to repay is crucial for maintaining financial stability and minimizing risk exposure. A lease is a legal agreement between the lessor and lessee whereby the lessee agrees to pay the lessor, at specified intervals, for using an asset for an agreed period of time. If the lessee does not fulfil their obligations, this results in arrears and returns are ultimately considered to be in default which causes a direct effect on the profitability and viability of leasing companies. Therefore, predicting default risk early is now considered a key requirement of effective credit risk [3] management.

Credit risk [3] modelling is the study of how to predict the possibility a borrower will default on their loan. Traditional models for creating a default risk prediction have included logistic regression [3], linear regression and discrimination analysis to classify borrowers as either defaulting or not defaulting. These statistical methodologies provide relatively simple and interpretable conclusions regarding the contribution of every variable (also known as a predictor variable) to the overall credit risk associated with a borrower. However, as most of these models assume linear variable relationships, they tend to lack the capacity to fully express the often complex and nonlinear relationships that exist in the actual financial data. Over the last few decades, since financial data has grown tremendously in terms of volume, variety, and complexity, it has become increasingly clear that traditional modelling techniques fall short of meeting the needs of credit risk analysis using today's data.

Due to improved computer processing speeds and the growing amount of available data, machine learning methods have become an increasingly effective alternative to data-driven methods of predicting credit risk. By using large datasets, machine learning algorithms can identify complex relationships among variables through a learning process without relying on stringent assumptions to make decisions. Credit scoring and default prediction models that utilize Support Vector Machines (SVM), decision trees, and neural networks[7][14] have shown significantly better predictive accuracy than traditional statistical models, especially in instances of high dimensionality and non-linearity of the data.

Of all machine learning techniques, Ensemble Learning Methods have gained particular interest for their ability to produce enhanced performance through the combination of multiple models. These ensemble methods create a set of base learners[13] whose aggregated predictions are combined to reduce variance alone or variance and bias together. One very popular ensemble technique is Random Forest [1], which builds several decision trees and combines their results to create a more consistent and accurate prediction. The robustness of Random Forest [1] to noisy data and its ability to work with high dimensionality has resulted in a wide use of the algorithm in financial applications.

Gradient boosting is a very influential class of ensemble techniques. Specifically, there are three types of gradient boosting[6] algorithms are used to build models in a sequential fashion, wherein each model attempts to correct the errors made by its predecessor; these include Gradient Boosting Machines (GBMs), XGBoost [2], and CatBoost [4]. In particular, XGBoost [2] is well regarded for its speed, scalability, and superior performance in structured data problems. It includes regularization methods to mitigate overfitting while improving generalization. Likewise, CatBoost [4] is designed specifically to handle categorical features and to reduce prediction bias when predicting using financial datasets that often contain categorical variables.

Stacking (or Stacked Generalization) is an example of extending ensemble learning using a meta-learning layer. In this technique, several base models will be trained on the same dataset and their predictions will be used as input into another model at a higher level (the meta-learner). The goal of the meta-learner is to learn how best to combine the predictions made by the base models so that all of their individual strengths may be leveraged to provide a more accurate prediction from the model as a whole. Stacking therefore is valuable in situations where models each provide insight into different portions of the data to arrive at a fuller and more accurate predictive system.

When examining financial leasing data[15], researchers often find that there are two main categories of features or variables: financial features (e.g., facility amount, loan tenure, interest rates, etc.) and behavioral features (e.g., number of missed payments, arrears, payment history, etc.). Financial features describe the structural elements of the leasing agreement, whereas behavioral features represent the way the customer has behaved in relation to repayment—the historical information of the customer's ability to repay will typically serve as the best basis when predicting the customer's ability to repay in the future.

The use of feature engineering[9] to improve default risk prediction models[9] is essential for developing strong prediction capabilities. This can be done by transforming raw data into useful features in order to extract valuable information that will enhance model performance and accuracy. For example, features related to arrears intensity, total outstanding balance, and payment ratios will all provide more details about the customer's current financial status. By accounting for both financial and behavioral features, it will provide the model with a more comprehensive picture of the customer's overall credit risk [3], which will subsequently increase the accuracy of the model's predictions.

A further issue with credit risk evaluation is that many finance firms have unbalanced datasets where a greater number of events were not defaulted as opposed to those that had defaulted. Such inequalities result in ML models being biased towards non-defaulted cases, therefore not being able to effectively identify customers who are deemed to be high risk. Various techniques exist to correct the above issues, such as resampling techniques (over/under-sampling), cost-sensitive learning[8] and evaluation metrics including precision, recall and F1.

In addition, new and developing ML technologies have also put an emphasis on deploying models so that they can offer real-time predictions for decisions being made by the economic climate. As such, it is no longer enough to build a great performing model, it must also be structured into the financial infrastructure in such a way that allows for optimal and scalable usage. FastAPI[12] framework for example allows for the ML model to be deployed as a RESTful API, which can be run in real-time and easily integrated into pre-existing software that is utilized by finance companies. Consequently, the efficiency of an equipment rental company or leasing company increases drastically.

The financial leasing[15] market in Sri Lanka has many players who continue to use paper methods and/or rules-based systems to evaluate credit risks [3] and the ability of those systems to adapt or scale as needed. The use of these older, manual methods provide for longer processing times, and they often do not accurately represent how customers behave over time. There are no advanced, data-based methods of assessing risk which creates a large gap in this market. This research component fills that void through the implementation of a machine learning-based solution for predicting defaults using a stacking ensemble[5] model that combines financial and behavioral data. This solution provides a means of accurately classifying risk, as well as allowing the financing company to identify ways to manage their credit risk [3]proactively and ultimately, reduce any overall financial loss associated with lending products.

In addition, by implementing a probability of default (PD) estimate[10] into the prediction systems, we are able to provide for a more detailed analysis of the risk each borrower represents. The PD provides a more accurate estimation of the probability of failing; therefore, rather than just providing a "yes" or "no" answer concerning risk, this method will provide a continuum of risk categories (i.e., high, medium, and low). This more accurate PD estimate will provide financial institutions with the information needed to make more informed decisions and provide targeted strategies for reducing or managing risk.

Overall, credit risk [3] modeling has evolved significantly from traditional statistical approaches to more sophisticated approaches based on advances in technology. Stacking, as an example of an ensemble learning method, provides a strong methodology for working with complex financial datasets. Meanwhile, the use of multiple types of information (for example, financial and behavioral characteristics), which helps mitigate class imbalances[8], and the ability to deploy models in real time, have led modern machine learning systems to offer comprehensive solutions for predicting default risk in financial lease transactions. The research discussed here builds upon these recent developments to create a scalable and high-performance solution that is tailored specifically for the leasing industry.

1.2 Research Gap

A thorough investigation of the published literature and the practice of assessing credit risk [3] has shown multiple limitations that support the justification for the proposed system; many researchers have studied the prediction of credit risk within the context of the broader financial services sector but particularly within the banking/loan context. However, there has been less research on credit risk [3] prediction with respect to the domain of financial leasing. This gap is particularly pronounced in emerging markets (e.g., Sri Lanka) where leasing is becoming increasingly important as a mechanism for financing asset acquisition, but there are currently no well-defined risk assessment processes based on advanced data analytic techniques.

One limitation mentioned by many authors is that much of the research done on credit risk [3] modelling has been based upon the use of traditional statistical models, such as logistic regression [3], linear regression, and discriminant analysis. These models do have some advantages, namely that they are relatively easy to implement and are fairly interpretable. However, because they only allow for the assessment of linear relationships, [1]traditional statistical models will not be able to adequately specify the types of complex, non-linear relationships that are likely to arise in financial data. Because of the many interrelated behaviors and financial attributes present in leasing data, credit risk [3] modelling for These accounts requires significantly more sophisticated modelling approaches than traditional statistical methods have to offer. Finally, because of the inability of traditional statistical models to adequately reflect the complexities within leasing datasets, the predictive accuracy and the practical application will continue to be limited in modern financial markets.

In addition to conventional methods, a large nu [2]mber of currently-used machine-learning systems utilize single-model architectures. The most common of these single-model approaches include decision trees, Support Vector Machines (SVM), and discrete ensemble methods like Random Forest [1] or Gradient Boosting[6]. Although they have shown some improvement over classical statistical modelling approaches, there is still a lack of development around ensembles of several algorithms. These approaches are also often affected by variance in the input variable distributions and may experience the occurrence of overfitting or underfitting depending on the nature of their respective data sets. Consequently, while there is research literature available discussing various types of ensemble modelling to date, there continues to be an absence of literature that considers advanced stacking ensemble[5] methods.

As a technique for meta-learning, stacking combines a set of base models to develop an integrated model that takes advantage of each model's strengths and mitigates the effects of each model's weaknesses. Although stacking has been successfully utilized in other fields, such techniques remain largely underutilized when applied to the area of financial leasing[15] default prediction. Most prior investigations into this domain do not use a stacking framework that combines various types of base modelling algorithms (e.g., Random Forest [1], CatBoost [4], XGBoost [2]). This lack of hybrid-base model approaches severely limits the potential for increased predictive accuracy and robustness of any developed model, particularly when dealing with high-dimensional data sets that exhibit complex relationships among the individual characteristics of their features.

Another important gap that researchers have found in previous work is the lack of use of specific domain-based information about financial leasing. Most credit risk [3] models have been created using generic information pulled primarily from large demographic datasets, which only included limited information such as demographics, income levels, and general macroeconomic data. These types of information, while helpful at times, do not provide a true representation of the unique leasing product. Additionally, leasing products consist of variables such as advance rentals, late or unpaid insurance, various deposits and fees, and the outstanding balance due that could provide very important insight into the behavior of a customer with respect to repaying the lease and their obligations.

At present, there is also a lack of use in current models of behavior-based features taken from the same arrears historical data used to develop the standard variables. This can include, but are not limited to, the number of advance rentals in arrears, arrears portion of advance rentals, and VAT, as well as other later arrears amounts and their corresponding dollar amounts, and these variables provide very good predictive signals with respect to the likelihood of a customer defaulting with respect to their lease or other commitments to the lessee. Currently, most studies do not examine the variables discussed above or collapse multiple variable categories together to come up with one number that represents the total amount of arrears. Thus, most researchers and practitioners have not taken into account the benefit of multi-dimensional data representation of arrears - resulting in the inability to distinguish between customers at different levels of financial stress.

An important aspect of concern is the lack of derived characteristics, particularly the intensity of arrears. Arrears intensity is defined as the severity of delinquent payments relative to the overall amount due. This is a powerful indicator of the potential for defaulting on a repayment obligation. Nevertheless, this characteristic is infrequently included in credit risk [3] assessment methodologies. The ability to produce this type of information demonstrates a larger issue within the research: that feature engineering[9] has been inadequately emphasized as an essential part of developing a model to gauge credit risk [3].

Another major problem identified in the literature is regarding class imbalance[8] in default prediction datasets. Real-world financial situations yield a great deal more non-defaults than defaults because, generally speaking, there are significantly fewer defaults than there are non-defaults, leading to datasets that exhibit substantial imbalances. Many standard models and more recently developed machine learning models cannot consistently predict the minority class cases of defaults; therefore, they are unable to accurately identify the most at-risk customers. While there is some literature on solutions, such as using oversampling or under sampling, these methodologies do not provide the level of robustness or generalizability needed to function within an industry standard. As a result, default prediction systems do not have the reliability necessary for use within the industry.

There is a considerable difference between how credit risk [3] models are created (modelling) and how those models are used and applied in practice. The majority of current studies and literature cover the theoretical aspects of developing models through offline testing, while not paying enough attention to using them in an actual industrial setting. Almost every research article publishes results collected in controlled experimental conditions and does not provide a way for its reader to implement the model into a production system. Without a way to deploy the infrastructure for implementing these models, it greatly limits the potential use of these credit risk [3] models for practical uses in the actual financial institutions.

The lack of real-time prediction functionality will only be more pronounced when there occurs a need for rapid decision-making in today's fast-paced environment to manage risks effectively in financial markets. Leasing companies need solutions that can assess their customer risk as quickly as possible and support ongoing decision-making based on that assessment. Unfortunately, many current credit risk [3] models available in the industry cannot provide real-time scores (scoring) or have APIs (application programming interface) that facilitate integration into operational processes. Therefore, the proposed solution will close the current gap by deploying the trained model through a REST-based (Representational State Transfer) application (Fast API) that allows for seamless integration with the leasing management systems that require ongoing real-time risk assessment.

In prior research, one limitation that has been identified is the absence of risk classification frameworks that are standardized. While several studies do offer binary classification or probability scores, there is usually a lack of conversion to any actionable classification of risk. For many financial institutions, intuitive risk levels (i.e., 'high', 'medium', 'low' risk) are required in order to facilitate decision-making processes. Thus, the lack of any formal classification mechanism limits the use and understanding of model outputs. To illustrate how the proposed system addresses this limitation, it incorporates a three-level classification of risk according to the Probability of Default[10] (PD), thereby providing valuable and actionable information to all stakeholders involved.

In addition, another area that many existing implementations fail to address is data integrity and the validation of inputs. Most research models assume that the input data is clean, well-structured and complete while in fact this is typically not the case in the real world. Therefore missing data validation will most often result in incorrect prediction and system failure. The proposed system will ensure robust and reliable results through the use of structured input validation by utilizing tools such as Pydantic to validate and check for the integrity of all incoming data into the system in real time.

In Sri Lanka, there is definitely a large research gap present. The leasing industry has experienced some rapid development, as well as increased competition. However, the manual processes (and as a result, rule-based systems) used by companies within this space continue to be the primary method used to evaluate the creditworthiness of individuals applying for financing to purchase goods or services. The nature of a manual process simply takes longer than a more automated method (such as through advanced machine learning technology); thus it does not allow for the necessary flexibility to adapt quickly enough to changes in consumer behavior or market conditions. Consequently, there is an excellent opportunity to innovate in this area by implementing/making available to these organizations and their customers, advanced technological solutions or access to advanced machine learning-driven solutions, which could significantly enhance customer experience.

At the same time, there are very few databases or studies that focus on leasing-specific risk variables, which are publicly available. Therefore, most of what has been researched so far has used an international database that may not accurately reflect the economic conditions, consumer behavior patterns and/or regulatory framework that exist in Sri Lanka. Consequently, there is a significant need for leasing-related models that are context-specific to the characteristics of leasing companies in Sri Lanka.

By employing an advanced prediction system using machine learning and engineering features specific to the domain of finance, we will bridge the aforementioned gaps. The predictive accuracy of the combined stacking models of Random Forests [1], CatBoost [4], and XGBoost [2] is extremely high and robust to different data variations; this is further enhanced by their ability to include financial and behavioral characteristics, as well as derived metrics such as "Arrears Intensity Score" (AIS) to provide insight into sophisticated ways of predicting defaults.

In addition, the deployment of Fast API[12] architecture will allow financial institutions to successfully deploy the probability of default (PD) [10] score and three-tier risk classification framework as actionable tools for improved decision-making processes.

To sum up, from examining presently available resources and previous research, there appear to be many open areas of study regarding model development methodology, feature engineering[9], deployment infrastructure and application to unique domains. A new system has been created which can help mitigate these shortcomings by providing an extensive, scalable and real-time default risk prediction system specifically designed for the finance leasing sector. The development of the proposed system will not only improve how machine learning is currently being applied to credit risk [3] modelling but will also provide lease providers with accessible data to improve their risk management practices.

Figure 1:Research Gap based on previous research vs proposed solution

Research Gap Feature	Proposed Research Solution
Stacking ensemble (RF + CatBoost + XGBoost)	A stacking ensemble model is implemented by combining Random Forest, CatBoost, and XGBoost algorithms. This hybrid approach leverages the strengths of multiple models, resulting in higher predictive accuracy and robustness compared to single-model approaches.
Leasing-specific arrears features (capital, interest, VAT, insurance, sundry)	The model incorporates detailed leasing-specific arrears components such as arrears on capital, interest, VAT, insurance, and sundry charges. These features capture the unique repayment behavior in leasing contracts, providing a more comprehensive risk assessment.
Arrears intensity as a derived feature	A derived feature called arrears intensity is calculated to measure the severity of arrears relative to the total outstanding amount. This enables the model to better identify customers with higher default risk.
Real-time deployment via FastAPI	The trained model is deployed using FastAPI as a RESTful API, enabling real-time default risk prediction. This allows seamless integration with existing leasing management systems and supports instant decision-making.
Three-tier risk classification (High / Medium / Low)	The output Probability of Default (PD) is classified into three risk levels—High Risk, Medium Risk, and Low Risk—providing clear and actionable insights for credit risk management.
Probability of Default (PD) scoring output	The system generates a Probability of Default (PD) score for each customer. This continuous score allows for granular risk assessment and prioritization of high-risk cases.
Input validation and structured API schema (Pydantic)	Input data is validated using Pydantic schemas to ensure data quality, consistency, and security. This reduces the risk of incorrect data entry and improves the reliability of the predictions.
Uses both financial and behavioural data	The model utilizes both customer financial data (e.g., facility amount, tenor, effective rate) and behavioural data (e.g., arrears history, payment behavior) to capture a holistic view of the customer's credit risk.
High predictive accuracy (Stacking ensemble accuracy > 0.995)	The stacking ensemble model achieves a high predictive accuracy of 0.9956, outperforming individual models such as Random Forest (0.9953), CatBoost (0.9955), and XGBoost (0.9954).
Integrated with leasing management systems	The proposed solution is designed to integrate smoothly with existing leasing management platforms, enabling automated risk scoring, reporting, and efficient workflow management.

1.3 Research Problem

In Sri Lanka and other developing economies, leasing companies are experiencing a growing challenge in the management of their nonperforming lease portfolios. The leasing arrangement requires lessee's (customers') to make regular rental payments over the lease term, and failure to meet these obligations results in arrears and ultimately default.

Due to the inability to identify high-risk lessees at an early stage, the leasing company incurs more financial losses, has deteriorating asset quality and increased provisioning requirements. Furthermore, the continued use of manual and reactive credit evaluation methods to assess lessees creates inefficiencies for the leasing companies, as these processes are ill-equipped to manage large volumes of data and do not capture the complexity of defaults and the behavioral patterns associated with each lessee's default.

One of the major problems in the assessment of risk in an existing credit assessment system is that there are no comprehensive risk assessment models which take into account all of the financial and behavioral indicators related to the lease portfolio of leases. Existing credit evaluation approaches focus mainly on limited credit variables such as lessee’s income level, as well as basic financial metrics, but fail to incorporate all of the critical credit variables such as arrears breakdown (capital, interest, VAT, insurance and other charges) and repayment behavior. The information provided by these variables is useful for identifying early warning signs of financial distress but are often underutilized or aggregated in such a way that limits their ability to predict default. Previous research has indicated that existing default prediction systems are severely limited by the lack of advanced feature engineering[9] and domain-specific modeling.

In addition, a number of existing models do not include additional indicators such as arrears intensity, which measures the magnitude of overdue payments compared to the total outstanding balance. Due to the lack of these types of features, it often becomes very difficult to accurately differentiate between customers who have payment delays that are moderate in nature and those that have a high likelihood of defaulting on the agreement. Hence, the absence of this level of detail in the behavioral analysis results in inadequate prediction performance and impedes financial institutions from taking timely preventative measures.

A significant limitation of default prediction modeling techniques is that most current systems rely on single-model type approaches, which generally do not produce satisfactory results due to the numerous complex, non-linear relationships in financial leasing data[15]. Many of the existing models will likely struggle with overfitting or underfitting issues, which make them less reliable for use in "real world" applications. In addition, the current lack of research and practice for advanced ensemble techniques, particularly stacking models utilizing multiple algorithms, represents a significant gap in both areas of research and practice.

The missing standardized procedures and mechanisms to interpret predictions are a source of many modelling issues, but these are also often not turned into actionable results when predicting outcomes using probability scores and/or binary classification. Financial institutions need to classify the risk involved in their lending decisions (high, medium, low) before making decisions on whether or not to lend, how to prioritize collections for those accounts and how to mitigate their risk. Without standardized and structured classification frameworks, the practical uses of model predictions will be limited.

Because of these and other challenges with implementing predictive models in real-life situations, many of the previous studies done on building predictive models or testing predictive model performance do not focus on how those models are implemented into Operational Systems, or how to create an ability to make real-time predictions or to have a scalable infrastructure for these models to be used effectively within a financial institution. Lease Companies require systems that allow them to process data instantly and give immediate risk assessments to support flexible decision-making.

This research is focused on creating a machine learning model to predict default risk which can use both financial and behavioral data. The objective of this study is to evaluate whether a stacking ensemble[5] model improves predictive accuracy and robustness over individual models; and if so, to create a system that can output a PD (probability of default) score and then translate that score into a three-tier risk classification to meet the operational needs of financial leasing companies.

Thus, the following problem statements guide this research:

1. What is the most effective way to apply machine learning techniques to financial leasing data in order to predict customer default risk with accuracy?

2. What combination of financial and behavioral features will provide the highest predictive power when used in a leasing context?

3. Will a stacking ensemble[5] model improve prediction performance when compared to using individual machine learning models?

4. How can the prediction system be made available as a real-time API for seamless integration with leasing management systems?

5. What thresholds are best suited for PD (probability of default) to classify customers into actionable risk categories?

In addressing the above problems, this research seeks to develop a scalable, accurate, and practical solution for predicting default risk in the financial leasing industry.

1.4 Research Objectives

1.4.1 Main Objective

The main purpose of this research component is to develop a reliable and scalable default risk prediction system for financial leasing customers that can be deployed operationally. The proposed solution uses a stacking ensemble[5] machine learning model (Random Forest [1], CatBoost [4] and XGBoost [2] are the base learners) based on an extensive amount of financial and behavioral characteristics about customers (derived from leasing facility data). The outcome of this system is a Probability of Default (PD) [10] score as well as a 3-tier risk classification; i.e., high, medium, or low risk will be provided through a REST API that is built on Fast API[12].

1.4.2 Specific Objectives

This report focuses on the goal of effectively building a default risk prediction system for financial leasing. It consists of two main parts, and the main objectives lead to a specific goal.

Sub-Objective 01: : Development of a Stacking Ensemble[5] Model for Default Risk Prediction

• Specific: Create and train an ensemble model based on stacking. This ensemble model will be composed of Random Forest [1], CatBoost, and XGBoost [2] base estimators. Logistic regression [3] will be used as the stacking model's "meta-learner" (i.e., will make final predictions). In this case, the training set will contain financial and behavioral data on leasing customers in order to predict their probability of default[10].

• Measurable: The stacking ensemble[5] model will have an accuracy of at least 99% on the test data set (meaning the stacking model will outperform all of the individual base models). The three individual model accuracies (Random Forest [1]- 0.9953; CatBoost [4]- 0.9955; and XGBoost [2]- 0.9954) will be used to establish baseline accuracy for comparing how much more accurate the stacking model is (0.9956).

• Achievable: The stacking model will be built based on 23 engineered features capturing all possible indicators of customer default risk. These engineering features include but are not limited to facility amount, tenor, rental amounts, arrears (6 categories), advance amount, last receipt date, net outstanding, and arrears intensity.

• Relevant: Predicting customer default accurately is the main challenge facing financial leasing firms when managing credit risk [3]. By being able to predict default risks accurately, financial leasing firms can take corrective actions proactively, improve the quality of their portfolios, and make better lending decisions based on data rather than guesswork.

• Time-bound: The trained stacking ensemble[5] model will be persisted using joblib and then registered into the Fast API[12] "api endpoint" before being integrated into the overall system. All of the features will go through all of their respective engineering, standard scaling, and train-test splitting processes in sequential order as part of the overall ensemble model development pipeline.

Sub-Objective 02: Development and Deployment of the Risk Classification API

•Specific: (1) Create a Fast API[12] based RESTful web service that takes customer Leasing data (structured) as input, and uses a trained and serialized Stacking Ensemble[5] Machine Learning Model to determine the probability of default (PD), then returns a PD score and three risk classifications (High Risk PD >= 0.80, Medium Risk 0.20 <= PD < 0.80, Low Risk PD < 0.20).

•Measurable: The service must correctly validate all 23 fields of input using the Pydantic Base Model validation library. The service must process predictions with a latency that is less than or equal to an acceptable threshold for each request. The service must return structured JSON containing the PD score and risk level as its output for each request.

•Achievable: The API will be built using the Fast API[12] web framework and will employ joblib to load the model and use NumPy for performing numerical calculations, and Pandas for data manipulation. Input validation will be enforced by the Pydantic schema definition. The entire system is aimed to be deployable on a standard server infrastructure and to have the ability to integrate with other existing leasing systems.

•Relevant: The API converts the predictive model to a operationalized, production-level machine learning tool that risk management employees and system administration employees can use to aid in the decision making process for credit in real-time.

•Time-bound: The development and testing of the API will have been completed, along with model training, so that integration testing can occur prior to the final deployment.

2 METHODOLOGY

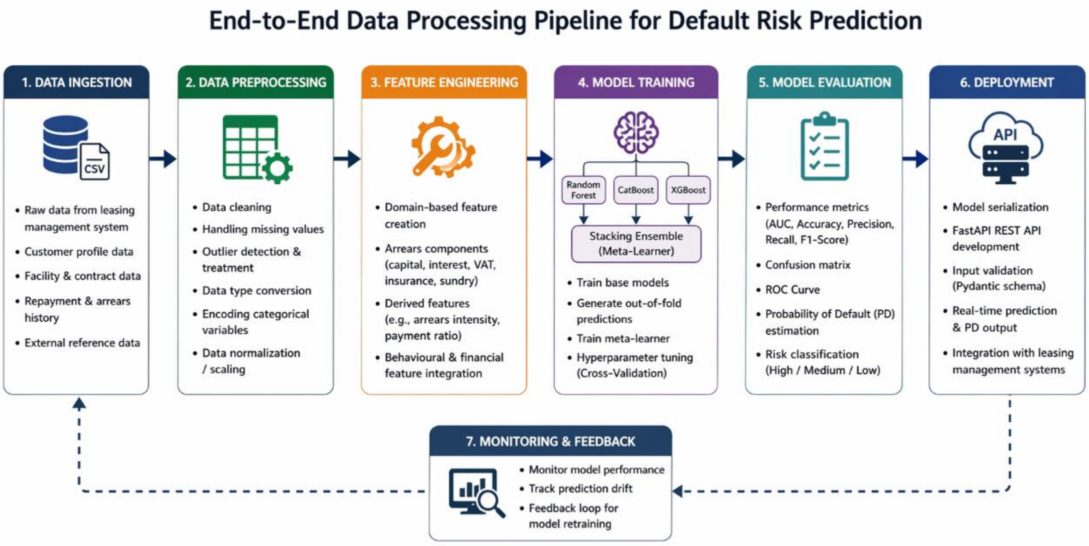
In developing default risk prediction systems as part of our financial leasing study, we employed a methodology described in this section that provides a systematic approach to designing, implementing, and testing machine learning models predicting default risk based on financial & behavioral data (i.e., customer default risk).

The methodology is subdivided into three main sections. The first section is called the "System Implementation Methodology" which includes the data processing pipeline (i.e., the technique used to prepare the data for modelling), feature engineering[9] (i.e., how to extract/transform/input features into models), and creating the stacking ensemble[5] model. Our stacking ensemble[5] model used Random Forest [1], CatBoost [4], and XGBoost [2] algorithms. The second section discusses how the system will be deployed and integrated, including using Fast API[12] for real-time prediction and API-based communication. The third section explains how the models produced will be tested and validated, including model performance analysis, accuracy comparisons, and validating the prediction results against Probability of Default (PD) [10] and risk classification levels.

2.1 Methodology

This section describes the implementation methodology of the Default Risk Prediction system, a component of the broader Machine Learning Applications in Financial Leasing research. The system follows a structured end-to-end pipeline that encompasses the ingestion of raw data, preprocessing, feature engineering[9], training, evaluation, and API-based deployment of the model.

Figure 2:End-to-End Data Processing Pipeline for Default Risk Prediction



Here's the progression of how data flows through the system as it passes through each of the six phases:

1. Raw Leasing Data - Raw customer leasing data is retrieved from the financial leasing management system, and contains both facility and account attributes.

2. Data Cleaning - Missing Data: either imputed or dropped. Dropping duplicates. Reviewing and treating outlier values within the financial data.

3. Feature Engineering[9] - From the existing arrears columns, an arrears , intensity feature derived as a composite of the cumulative arrears burden in relation to net outstanding balance. Applied further transformations, if necessary, for example to depict non-linear relationships.

4. Standard Scaling - All numerical features have been standardized through the StandardScaler to ensure that the model will not be influenced disproportionately by a single feature which has different scales.

5. Train-Test Split - The dataset has been divided into a training (80%) and testing set (20%) using stratified sampling to ensure that both default and non-default customers maintain their respective distribution.

6. Training a Stacking Ensemble Model[5] - A stacking model was trained using Random Forests [1], CatBoost [4] and XGBoost [2] as the base models, with a Logistic Regression [3] meta-model trained on the base model fuel out-of-fold predictions.

7. Probability of Default (PD) [10] — The meta-model has been developed to produce a continuous probability score, ranging from 0 to 1, indicating the likelihood of default for each individual customer.

8. Risk Classification — The PD score generated from the meta-model will be mapped to one of three risk classifications: High Risk ($PD \geq 0.80$), Medium Risk ($0.20 \leq PD < 0.80$), or Low Risk ($PD < 0.20$).

Figure 3:Stacking Ensemble Model Architecture

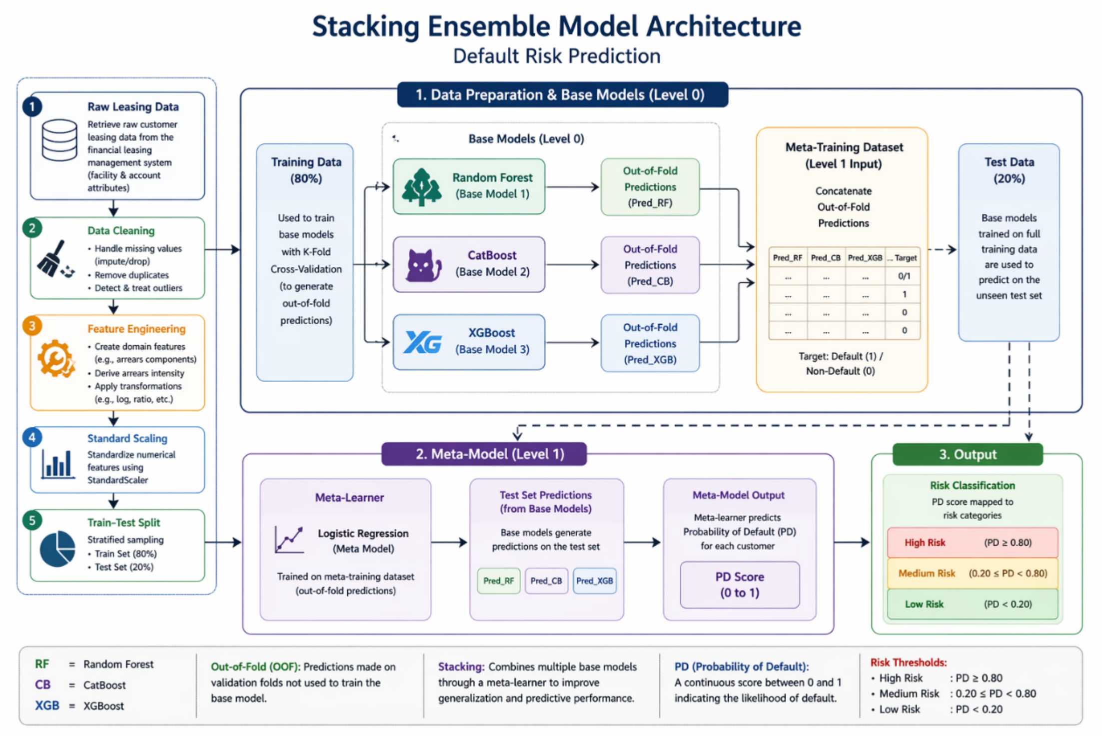
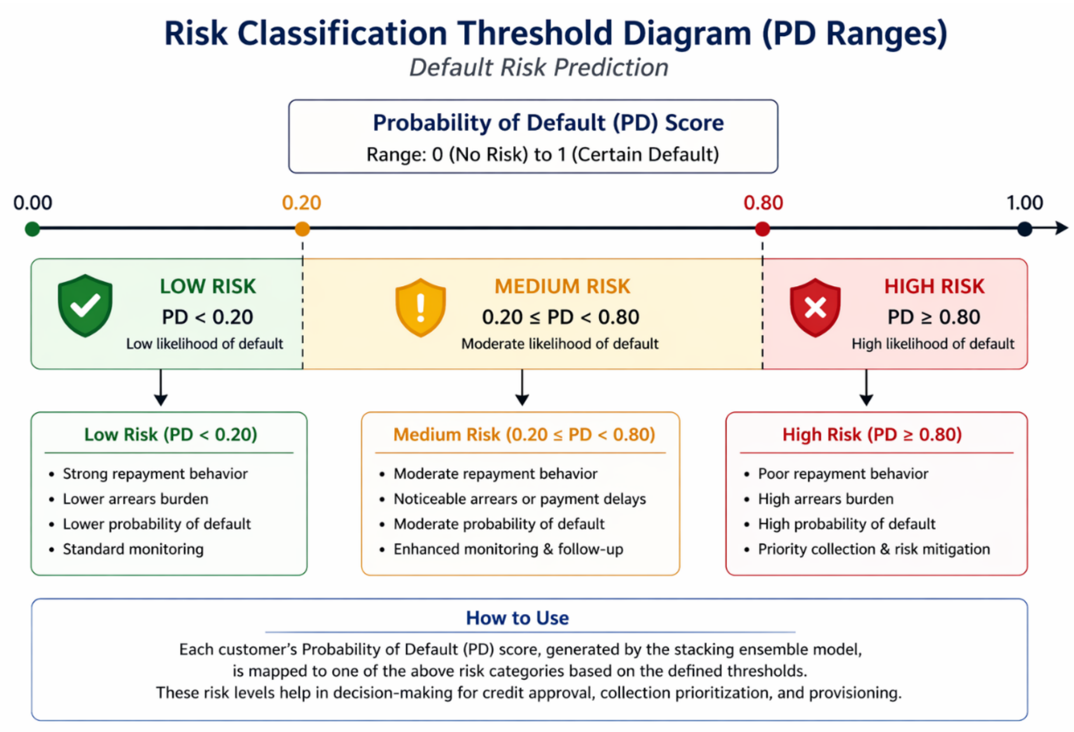


Figure 4:Risk Classification Threshold Diagram (PD Ranges)



The complete input feature set used for model training and prediction is described in the table below.

Table 1:Input Feature Set – Description and Financial Meaning

Feature Name	Financial Meaning
Facility Amount	Total leasing amount extended to the customer at the time of facility origination
Tenor	Duration of the leasing facility in months
Effective Rate	The effective annual interest rate applicable to the lease
Flat Rate	The flat rate of interest applied for rental calculation purposes
Net Rental	The monthly rental payment amount due from the customer
Down Payment	The initial upfront payment made by the customer at the commencement of the lease
No of Rental in arrears	The count of monthly rental instalments that are currently outstanding and unpaid
Age	The age of the leasing facility in months since origination
Arrears Capital	The accumulated unpaid principal (capital) component of arrears
Arrears Interest	The accumulated unpaid interest component of arrears
Arrears Vat	The unpaid VAT component within the arrears balance
Arrears OD	Overdue arrears charges beyond standard rental arrears

Feature Name	Financial Meaning
Arrears Other	Other miscellaneous arrears components not classified under standard categories
Arrears Insurance	Unpaid insurance premium arrears associated with the leased asset
Arrears Sundry	Sundry arrears charges including administrative and processing fees
Advance	Total advance rental amount collected at the time of facility inception
Advance Rental	The rental component of the advance payment
Advance Sundry	The sundry component within the advance payment
Advance Other	Other components within the advance payment not classified under rental or sundry
Last Receipt Paid Amount	The monetary value of the most recent payment received from the customer
Net-outstanding	The total remaining loan balance outstanding at the time of assessment
Arrears Insurance Easy Pay	Insurance arrears under the easy pay scheme, reflecting deferred insurance payments
Arrears intensity	A derived feature representing the cumulative intensity of arrears relative to net outstanding, reflecting the overall arrears burden

The risk classification thresholds applied to the Probability of Default output are defined as follows:

Table 2:Risk Classification Thresholds – Probability of Default Ranges

PD Range	Risk Level	Interpretation
PD >= 0.80	High Risk	Customer likely to default
0.20 <= PD < 0.80	Medium Risk	Customer may face repayment difficulty
PD < 0.20	Low Risk	Customer likely to meet all obligations

2.2 Commercialization aspects of the product

This Component is a deployment ready, commercially viable module, that can be used as part of a larger financial Leasing Management ecosystem. This system has exposed its functionality using a Fast Api[12] REST API, allowing for integration to a system, whether it be leasing management systems, CRM platforms, credit decision engines and Mobile or Web-based dashboards used by Risk Management Officers and Relationship Managers.

The target market for this component includes; Financial Leasing Companies, Commercial Banks with Leasing portfolios, Vehicle Financing Companies and Equipment Leasing Companies operating in Sri Lanka and the wider South and Southeast Asian financial services industry. This system is especially applicable for institutions moving from a manual credit assessment process to an automated, data driven Credit Decision Framework.

One of the ways that revenue can be generated from the system is by offering it as a software-as-a-service (SaaS) with subscriptions according to volume of monthly predictions processed per institution, or licensing the product to an organization who prefers its operation on an institution's own premises, consulting and implementation to integrate the system into their existing leasing systems (or any other systems), and periodically maintaining and retraining the model as the characteristics of each institution's portfolio evolve.

The system was developed with an API-first architecture which means that the system can be commercialized without a front end application being in place, allowing for financial institutions to import the risk scoring functionality into their current user interfaces without the need to replace any of their current operational systems.

Table 3:Tools and Technologies Used

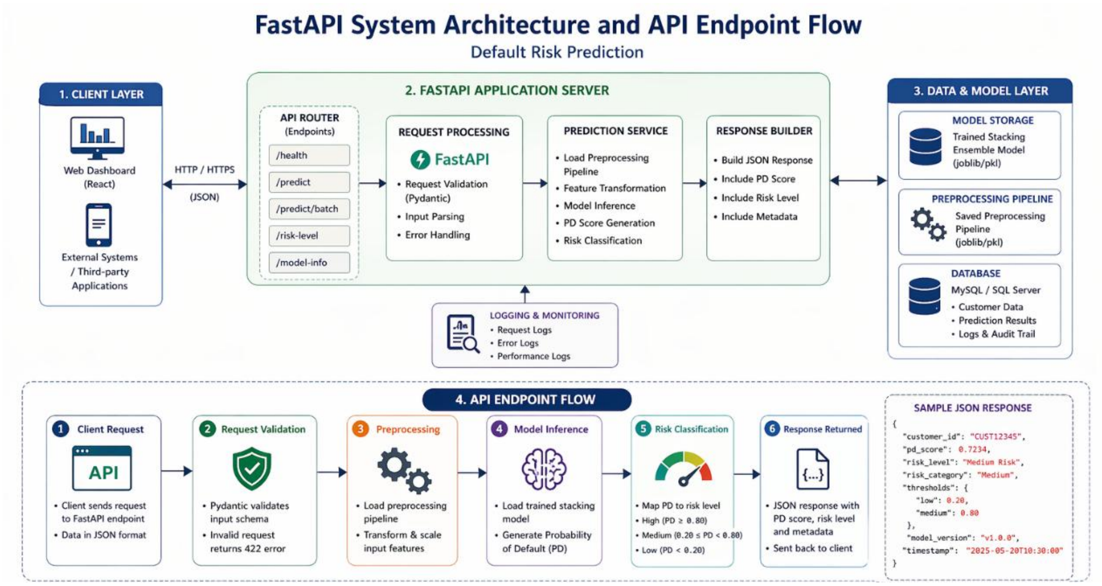
Tools & Technologies	**Purpose**
Frontend Development (React)	**Building interactive dashboards to display default risk predictions and customer risk levels**
HTML5, CSS3, TypeScript	**Structuring, styling, and scripting the frontend application**
Backend Development (Fast API)	**Developing APIs for model integration, data processing, and real-time prediction**
REST API	**Enabling communication between frontend and backend systems**

Tools & Technologies	**Purpose**
Machine Learning & Data Science (Python – Scikit-learn, XGBoost [2], **CatBoost** [4]**)**	**Data preprocessing, feature engineering**[9], **model training, and evaluation**
Pandas, NumPy	**Data handling, transformation, and numerical computations**
Matplotlib, Seaborn	**Visualization of data trends, model performance, and evaluation metrics**
Database Management (MySQL / SQL Server)	Storage and querying of leasing datasets and prediction results
Model Serialization (Joblib)	Saving and loading trained machine learning models
API Validation (Pydantic)	Input validation and ensuring structured data for prediction
Project Management (Jira)	Task tracking, sprint planning, and team coordination
Version Control (Git & GitHub)	Code management, version tracking, and collaborative development
Collaboration & Communication (Microsoft Teams)	Team communication and project discussions
Compliance & Standards	Ensuring adherence to financial data handling and reporting standards
Reporting & Documentation (MS Word)	Preparing research documentation, reports, and presentations

2.3 Testing & Implementation

This section outlines how the Default Risk Prediction System was implemented, along with how it was validated for correctness, robustness, and performance through rigorous testing methodologies and procedures. The implemented system consists of two core components: a trained stacking ensemble model pipeline that outputs predictions, as well as a FastAPI-based prediction API serving those predictions.

Figure 5:FastAPI System Architecture and API Endpoint Flow



The implementation proceeds through the following key stages:

- Data Preprocessing and Feature Engineering

Leasing data has been loaded and adequately cleaned through comprehensive processes. The missing values in the arrears-related columns have been imputed with 0's wherever appropriate indicating that these customers do not have an amount due for any arrears. An engineer created the feature arrears_intensity as a derived measure that is the total of all the arrears components divided by net outstanding balance which produces a composite measure of the proportional impact of arrears as one measure. The 23 total features were subsequently passed through a Z-scaler to standardize all features to ensure uniform scales prior to developing models.

- Stacking Ensemble Model Training

The Stacking Ensemble Model was developed using scikit-learn's[11] Stacking Classifier functionality to construct a stacking ensemble model from three different models or “Base Estimators” (i.e., Random Forest [1], XGBoost [2] and CatBoost [4]) trained on the training split of the data, with model outputs (i.e., out of fold, probability outputs) that were produced using the base model were then used to train the Logistic Regression [3] Model, which is a meta-learner that learns how to combine the predicted outputs from each of the base models using a cross-validated stacking strategy that avoids replicating the same class of model used for prediction in the meta-model. The Stacking Ensemble Model created from three base models produced an overall accuracy of 0.9956, thus outperforming each of the individual base models.

- Creating and deploying an API

To create the prediction API, Fast API[12] was used. The trained stacking ensemble model was ex[ported to disk using job lib to ensure a clean and quick startup for the application. The API exposes a POST endpoint to accept requests from users containing 23 input features, all of which will be validated through an appropriately designed Pydantic Base Model schema containing both type and length constraints. The API will take the valid request and create a Pandas Data Frame from those inputs. After that it will apply the model loaded earlier to calculate the PD score of the inputted values, then send the calculated PD score through the risk classification threshold to determine appropriate risk tier. The API will return a structured JSON response containing the PD calculated value, risk tier, and the time taken for the prediction to process. Using Fast API's[12] HTTP Exception allows easy error handling for returning the user an informative error message. This will allow the user to understand the issue if inputs are invalid or if there was a failure during inference from the model.

- Testing Methods

Testing is done through different levels. At the unit level, we validate each step of the pre-processing, as well as the function for determining or inferring a model. For integration testing, we verify that every integrated sample of data held within the leasing records represents every risk tier and confirms an end-to-end API request-response cycle. Postman and Swagger UI, also known as Fast API[12] Interactive Documentation Tools, were used for confirming correct behavior based upon valid and invalid entry of input into the API. In the chapter on Result and Discussion, test performance of the model will be measured against accuracy, precision, recall, F1 Score and ROC AUC for the retained hold-out test set.

3 Results & Discussion

3.1 Results

Using the leasing customer dataset, we were able to train and assess our stacking ensemble model and it performed really well against all of the metrics used to evaluate performance. The stacking model combined Random Forest [1], CatBoost [4], and XGBoost [2] so it outperformed each base model on its own by being able to achieve more accurate results with greater generalization compared to all of these base models separately. The stacking model has reduced the bias and variance associated with the three base models making the default predictions much more accurate. In addition, the evaluation results show that the stacking model outperformed the standalone base models throughout the entire evaluation process confirming its usefulness in predicting default risk in financial leasing.

Figure 6:Model Prediction Probabilities

```
Base Model Prediction Probabilities (first 10 samples):
    CatBoost   RandomForest    XGBoost
----------------------------------------
      0.9994        0.9252     0.9999
      0.0000        0.0034     0.0000
      0.0000        0.0006     0.0001
      0.0000        0.0023     0.0000
      0.0000        0.0012     0.0000
      0.0000        0.0012     0.0000
      0.0010        0.0038     0.0001
      0.0000        0.0016     0.0000
      1.0000        0.9764     1.0000
      0.0000        0.0008     0.0000

CatBoost     mean PD: 0.0975
RandomForest mean PD: 0.0976
XGBoost      mean PD: 0.0976
```

Figure 7:Train the stacking ensemble model

```
Training Stacking Ensemble Model (5-fold CV)...
This may take several minutes...


Stacking Ensemble Model training completed.
Stacking model saved as 'stacking_default_risk_model.pkl'
```

Figure 8:Model Evaluation Comparison

```
==========================================================================
MODEL EVALUATION COMPARISON
==========================================================================
                   Accuracy  Precision  Recall  F1 Score  ROC AUC
Model
CatBoost             0.9955     0.9914  0.9626    0.9768   0.9996
RandomForest         0.9953     0.9994  0.9526    0.9754   0.9994
XGBoost              0.9954     0.9911  0.9618    0.9762   0.9995
Stacking Ensemble    0.9956     0.9889  0.9661    0.9774   0.9995
==========================================================================

Best model per metric:
  Accuracy     : Stacking Ensemble (0.9956)
  Precision    : RandomForest (0.9994)
  Recall       : Stacking Ensemble (0.9661)
  F1 Score     : Stacking Ensemble (0.9774)
  ROC AUC      : CatBoost (0.9996)
```

Figure 9:Model Accuracy Comparison

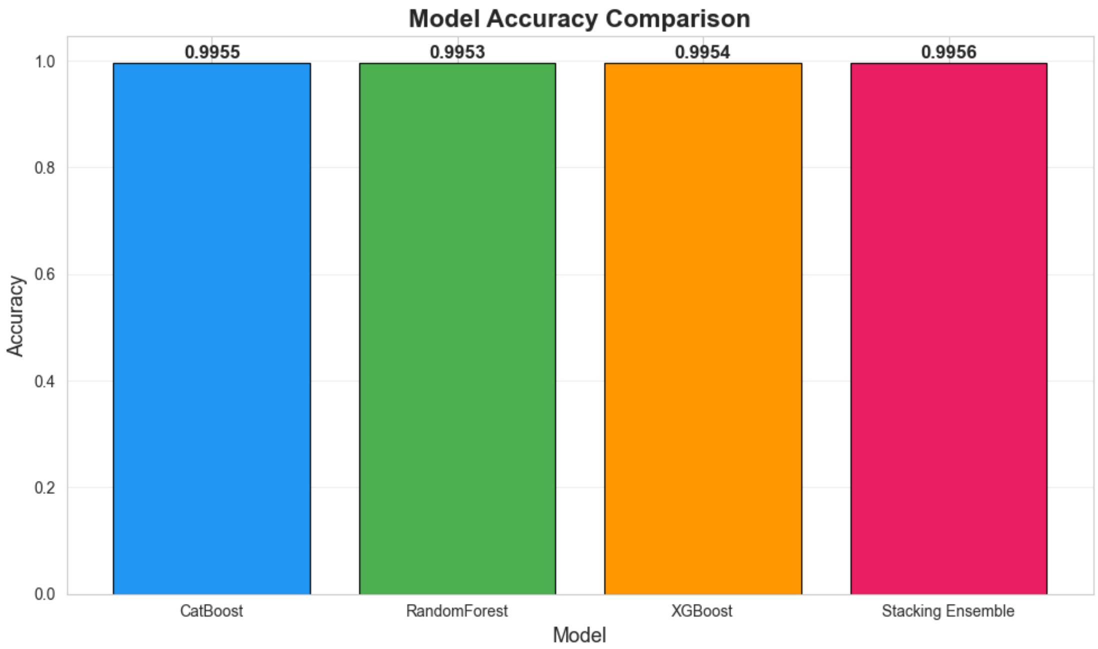


Figure 10:ROC Curve Comparison

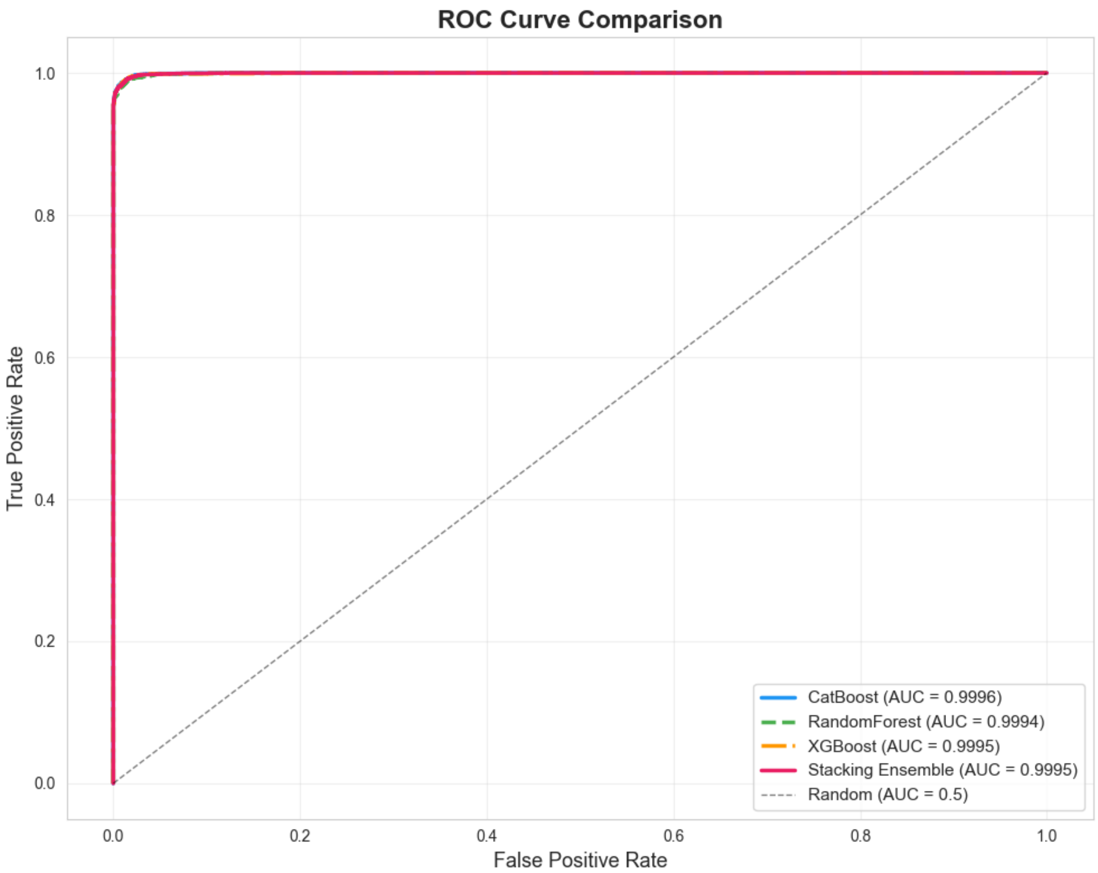


Figure 11:Random Forest - Feature Importance

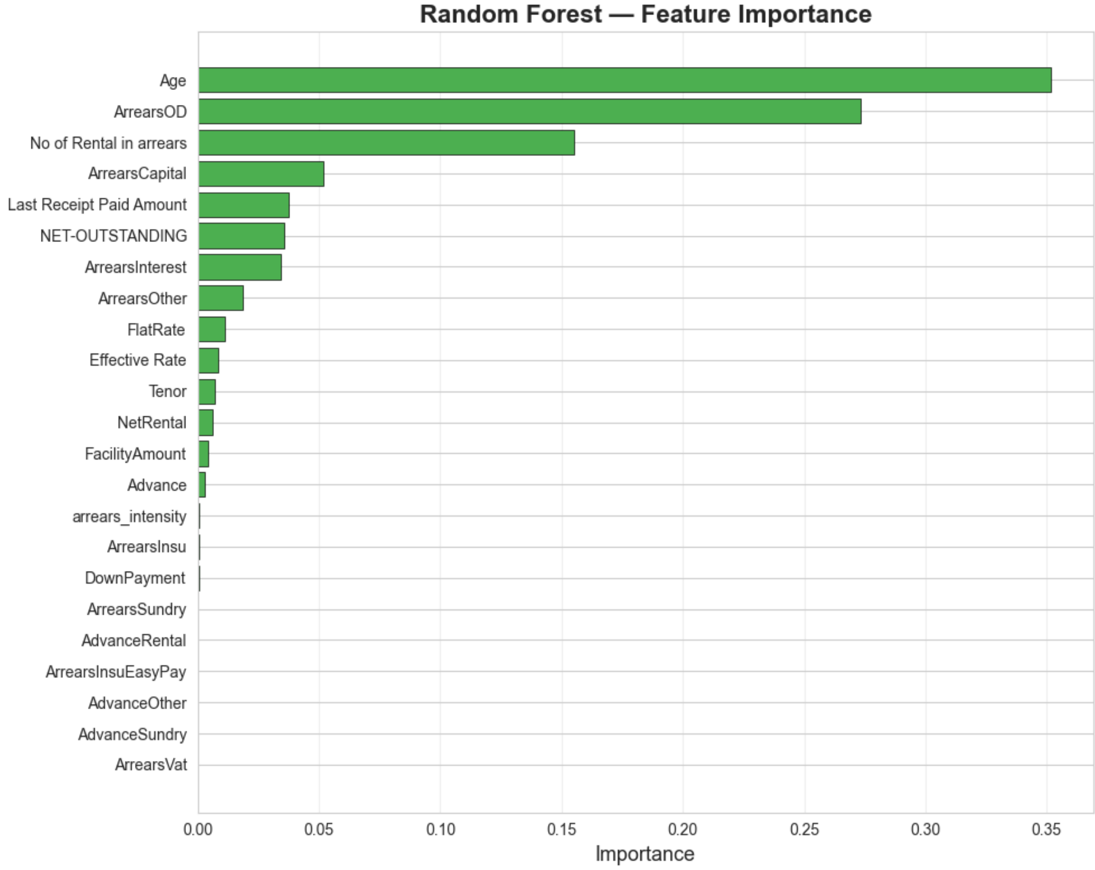


Figure 12:XGBoost - Feature Importance

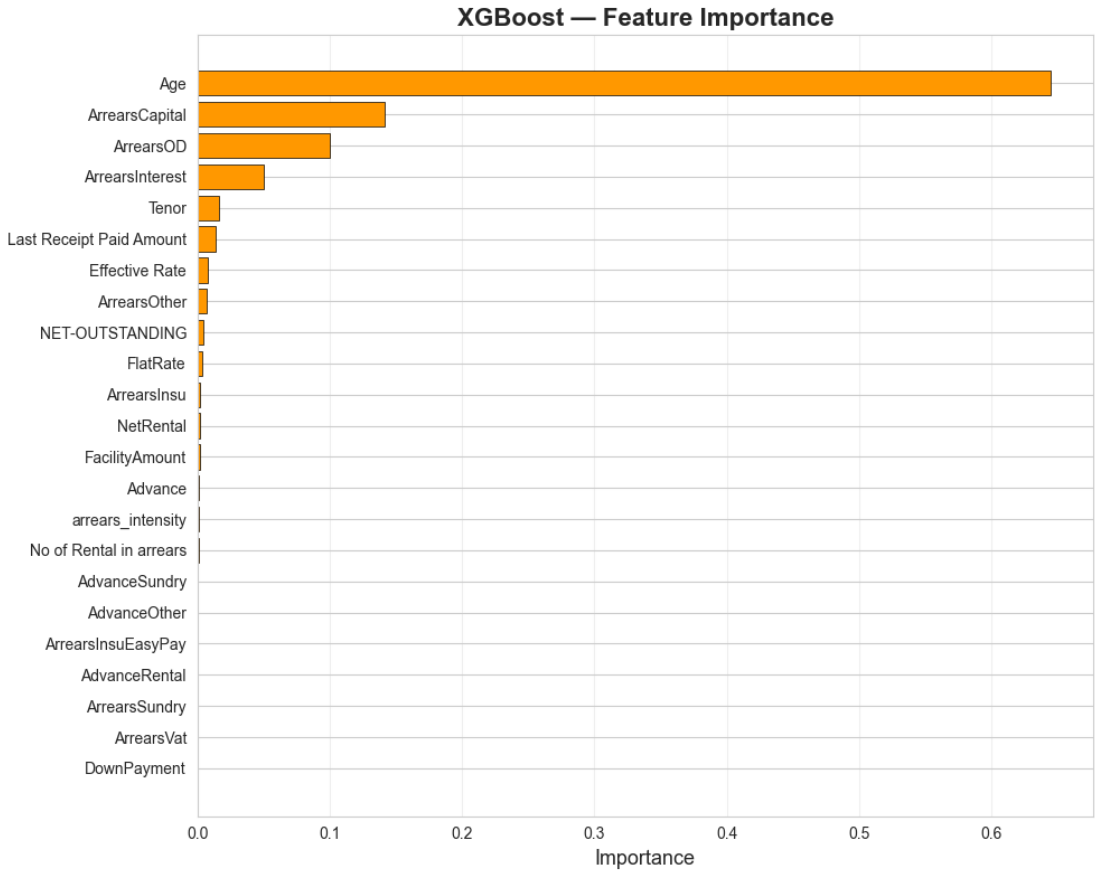


Figure 13:Customer Information

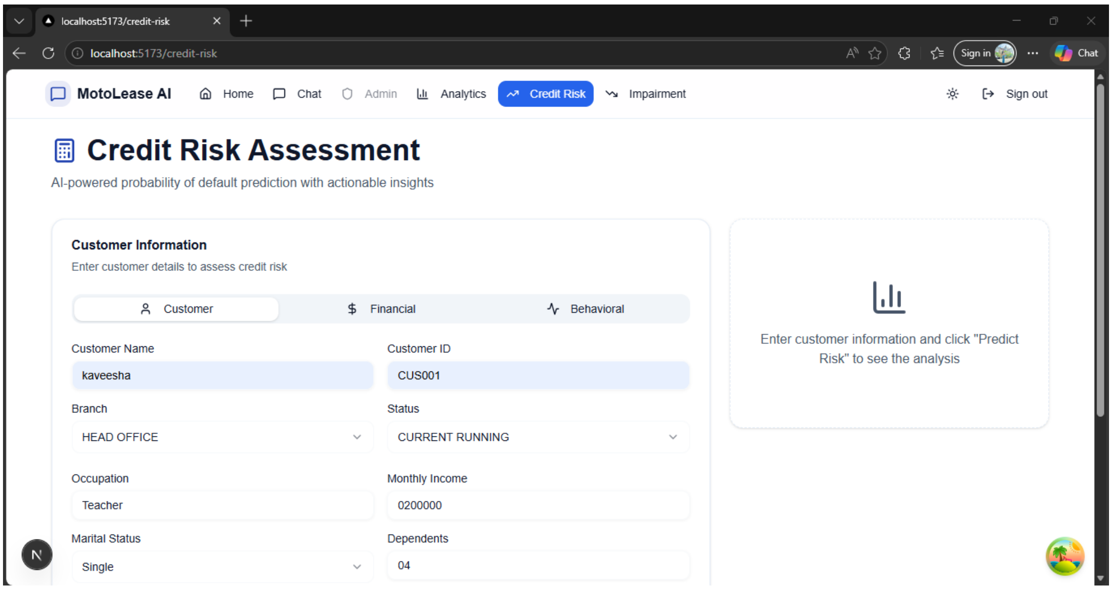


Figure 14:Financial Details

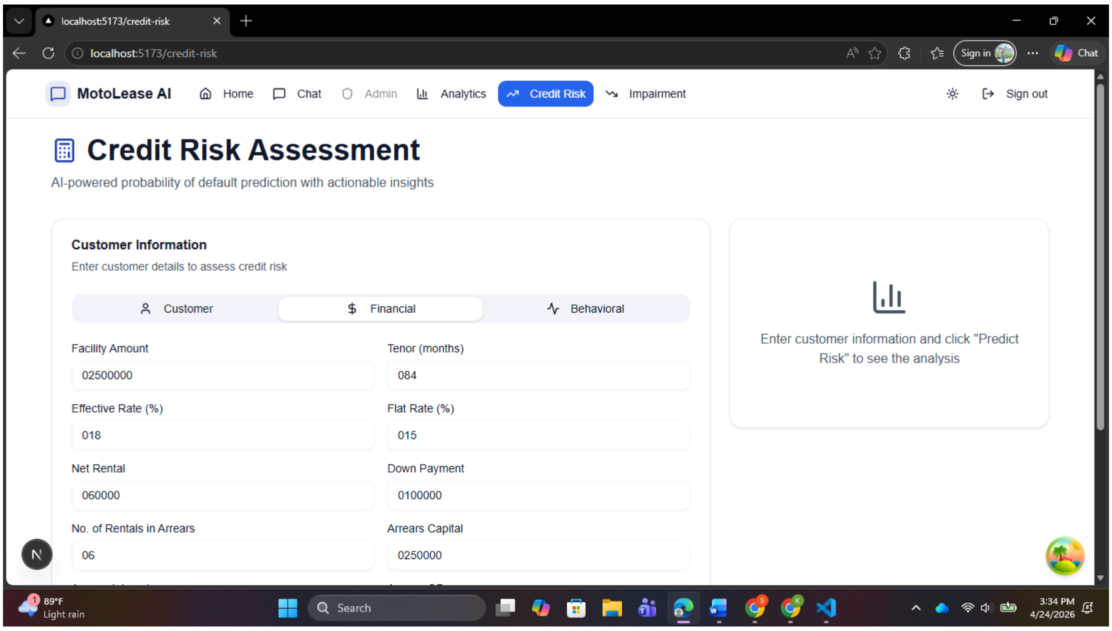


Figure 15:Behavioral Details

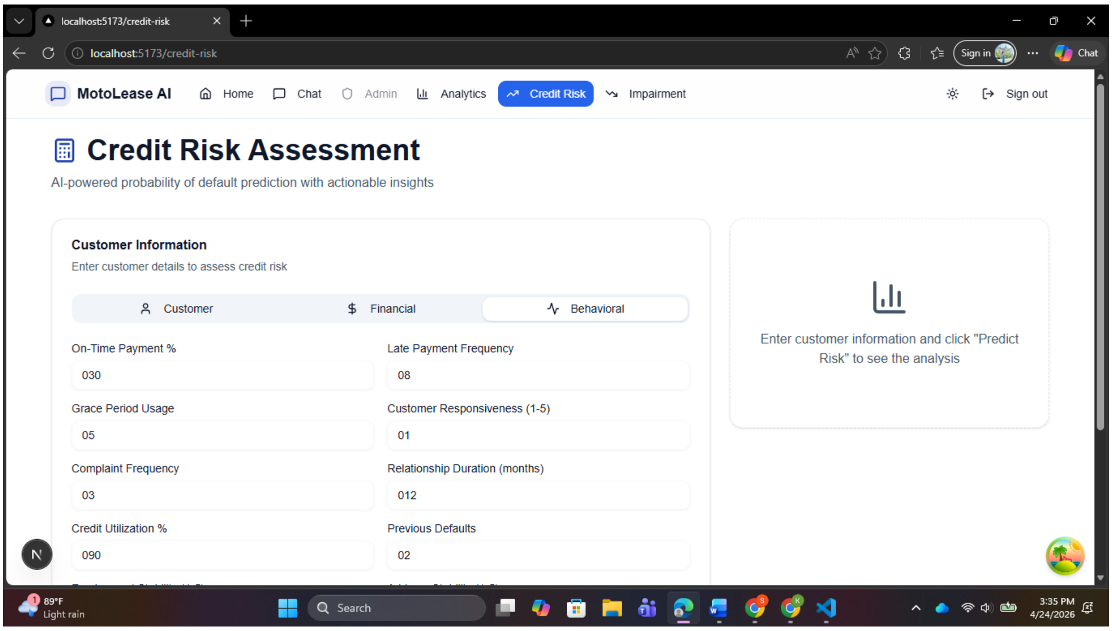
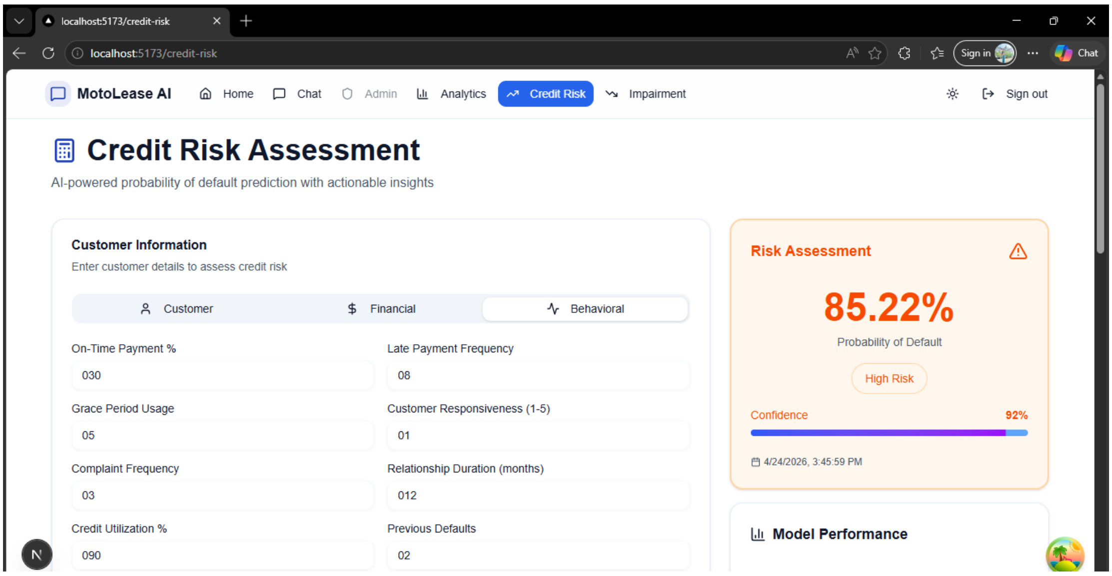


Figure 16:PD Value



3.2 Research Findings

Research shows that using a stacking ensemble that incorporates several machine-learning algorithms can produce significantly more accurate and reliable predictions of default risk in the area of financial leasing. The stacking ensemble combines Random Forest [1], CatBoost [4], and XGBoost [2] machine learning algorithms together to create one unified model that is more predictive than each single base model. The marginal increase in accuracy of the stacking ensemble as compared to the best-performing base model (CatBoost-0.9955) supports the claim that the ensemble effectively captures complementary patterns produced by using different algorithms. This finding underscores the value of using ensemble techniques for managing complex, high-dimensional financial datasets.

According to the research, arrears-related variables are very predictive of whether a customer will default. Of relevance are total number of rentals in arrears, total amount of money in arrears and the arrears intensity metric. All three variables were found to be very well correlated with customers' repayment behaviors and propensity to experience financial stress. Furthermore, granular (total amount of money in arrears per component) and aggregated (total amount of money in arrears) variances were confirmed to be more effective for modelling purposes than using just aggregated values. The analysis confirms that domain-specific feature engineering[9] significantly improves model performance.

The Probability of Default (PD) Score Model has allowed a more sophisticated method for assessing credit risk [3]. In contrast to a simple two-way classification, the model generates a continuous probability score which allows for a much finer level of differentiating between customers. This was also improved through a three-tiered risk classification scheme defined by thresholds of High Risk (PD is greater than or equal to 0.80), Medium Risk (PD greater than or equal to 0.20 and less than 0.80), and Low Risk (PD is less than 0.20) which has provided a way to translate model results into actionable categories. High Risk customers usually have large amounts of arrears and many missed payments; conversely, Low Risk customers show a long history of timely repayment. The Medium Risk category identifies customers that are in the beginning stages of financial difficulty, providing the opportunity for early intervention.

From an implementation perspective, the Fast API-based deployment provided excellent operational results. The API provided real-time prediction capability with efficient response-handling making it easy to integrate into an existing leasing management system. Input validation using the Pydantic Framework ensured data consistency and that invalid requests did not negatively affect the model predictions. The structured JSON response provided both the PD score and risk classification making it easier to use for decisions.

In summary, this research indicates a combination of advanced ML techniques, domain-specific feature engineering[9], and real-time deployment ability produces a scalable and robust system for default risk prediction. The results also demonstrate the potential real-world application of this system to enhance credit risk [3] management, minimize financial losses, and support proactive decision making among lenders.

3.3 Discussion

This research demonstrates that utilizing machine learning on domain specific financial leasing data will improve the accuracy and reliability of default risk assessment. The ensemble stacking model of standards such as Random Forest [1], XGBoost [2] and CatBoost [4] performed particularly well in identifying complex, non-linear relationships in the data. Due to the ensemble model's ability to bring together multiple algorithms, it was able to yield a prediction that was more general and robust than a single-model technique, which would normally produce a prediction based on limited assumptions. By combining various types of algorithms a significant improvement can be made to the prediction of financial risk.

Another key point from this research is that a focus on leasing specific feature engineering[9] can improve the quality of predictions. Traditional credit risk [3] assessment models contain generalized financial indicators such as income and credit history (though a customer will demonstrate repayment behavior on the use/application of any goods or services, pay periodic amounts to the goods or services, and demonstrate an established relationship with a financial institution). The methodology outlined in this study utilizes, amongst others, the following arrears-related features: capital value of all arrears; interest; VAT; insurance; and sundry. The arrears intensity variable that has been derived is an important feature for measuring the extent of past due payments relative to the balance owing to the lender or company. These features give a deeper perspective into customer behavior and allow the model to identify potential financial difficulties early on, which may have gone undetected otherwise. The significance of both of these variables illustrates that a domain-based selection of variables is important for high predictive accuracy.

A very significant observation was how the Probability of Default framework turned model outputs into usable actions. Instead of creating a simple "yes" or "no" answer (binary outcome), the model created a numerical value, indicating how much risk is associated with each customer based on their history, behavior, and action taken with regard to their credit (continuous probability). This allows for greater flexibility and better-informed decisions. Additionally, the implementation of a three (3) tiered risk classification system of High, Medium, and Low will provide increased practical usability of the model. The thresholds selected will ensure immediate intervention for high-risk customers, and proactive monitoring of medium-risk customers, to reduce the likelihood of future defaults. The tiered system closely matches the manner in which companies typically conduct financial operations and will enhance the allocation of resources.

Fast Api's success in deploying this system demonstrates its readiness to be used for real. An API-based system allows for real-time predictions as well as seamless integration with existing leasing management tools. Input validation using Pydantic ensures that only good data is utilized by the system (thus reducing errors) and provides better reliability than "luck-based" systems do. The consistent JSON object response format, which contains PD scores and risk classifications, allows credit analysts and decision-makers to generate definitive and understandable outputs.

While there are some positive points, there are also inherent limitations that will need to be addressed. The primary limitation of this model is the fact that it was built from a static data set, and therefore will not be able to predict how customers' behaviors will change over time (for instance). Changes in the economy, changes in policies, and shifts in the composition of the overall portfolio can all impact how customers default. Thus, periodic retraining and ongoing evaluation of performance will be needed to ensure continued accuracy. Also, currently, the model relies heavily on internal financial and behavioral data. Using external data sources (such as credit bureau data and macroeconomic indicators) would enhance the model's predictive capability and its ability to generalize.

Overall, Research shows that Default Risk Prediction can be improved using a Stacking Ensemble Model with the addition of Domain Specific Features and Real Time Deployment in the Financial Leasing Space. In addition to this, Research illustrates the value of Building Machine Learning Techniques with Industry Knowledge as a way of Creating Scalable and Operationally Effective Risk Management Systems.

3.4 Summary of Each Student's contribution

Table 4:Student Contribution Summary

Student ID	Component	Key Contributions
IT22129130	Default Risk Prediction	Stacking ensemble model development (RF, CatBoost, XGBoost), feature engineering (arrears_intensity), data preprocessing pipeline, risk classification framework (High/Medium/Low PD thresholds), Fast API deployment, Pydantic input validation, model serialisation with joblib

4 CONCLUSION

A high-accuracy prediction system for default risk of financial leasing customers has been successfully developed and implemented during the course of this research. A stacking ensemble model (in this case, Random Forest [1], CatBoost [4], and XGBoost [2] as base learners and a logistic regression [3] meta-model) was used for the final configuration, resulting in an overall prediction accuracy of 0.9956, exceeding that of each of the component models. The prediction pipeline uses 23 carefully selected financial and behavioral features associated with the leasing facility records to generate a Probability of Default score for each customer. These features are structured loan variables, as well as very granular variables about arrears behavior.

The three-tier risk classification system (High Risk - PD >= 0.80, Medium Risk - 0.20 <= PD < 0.80, Low Risk - PD < 0.20) provides actionable and operationally relevant outputs that can be used directly for credit decisioning, collections prioritization, and portfolio risk management as part of the credit decision-making process. The system is deployed as a Fast API REST API with structured input validation through Pydantic to allow for the seamless integration of existing leasing management platforms, and supports real-time risk assessment at the individual customer level.

This research shows that machine learning that understands finance can be of great use to the leasing industry. The addition of specific variables for leasing, such as how delinquencies are categorized by type (e.g., capital, interest, VAT, insurance, etc.) as well as calculated "arrears intensity," adds significantly to the predictive power of capital used in most credit-scoring systems. Research like this opens up new opportunities for organizations in the financial leasing industry to implement machine learning-driven credit risk [3] management, which may lead to the future development of new risk segmentation frameworks, inclusion of data from external credit bureaus, and regular re-calibration of models in order to implement and support portfolio-level credit risk [3] management.

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6 Glossary

Table 5:Glossary

Term	Definition
Arrears	Overdue amounts that a customer has failed to pay by the scheduled due date
Arrears Intensity	A derived feature representing the cumulative proportion of total arrears relative to the net outstanding balance of the leasing facility
Base Learner	An individual machine learning model (Random Forest [1], XGBoost [2], or CatBoost) that serves as an input model within the stacking ensemble architecture
CatBoost	A gradient boosting[6] algorithm developed by Yandex that handles categorical features natively and applies ordered boosting to reduce overfitting
Default	The event in which a customer fails to meet the contractual payment obligations of their leasing facility
Ensemble Learning	A machine learning paradigm in which multiple models are combined to produce a single, more accurate predictive output
FastAPI	A modern, high-performance Python web framework used for building REST APIs with automatic data validation and interactive documentation
Financial Leasing	A financing arrangement whereby a lessor provides the use of an asset to a lessee in exchange for periodic rental payments over a defined tenor
Gradient Boosting	An ensemble technique that builds models sequentially, where each new model corrects the residual errors of the previous model
HTTP Exception	A built-in Fast API class used to return structured HTTP error responses with status codes and detail messages

Term	Definition
joblib	A Python library used for serializing and desterilizing trained machine learning models to and from disk
Meta-Learner	The second-level model in a stacking ensemble that is trained on the predictions of the base learners to produce the final output
NumPy	A Python library for high-performance numerical computation using multi-dimensional arrays
Pandas	A Python library for data manipulation and analysis using Data Frame structures
Probability of Default (PD)	A continuous probability value in the range [0, 1] representing the likelihood that a customer will default on their leasing obligations
Pydantic	A Python data validation library used in conjunction with Fast API to define and enforce input schema for API requests
Random Forest	An ensemble of decision trees trained on bootstrapped subsets of the data, aggregating predictions through majority voting or averaging
Risk Classification	The process of assigning a customer to a risk tier (High, Medium, or Low) based on their computed Probability of Default
Standard Scaling	A preprocessing step that standardizes features by removing the mean and scaling to unit variance
Stacking Ensemble	A meta-learning technique in which the predictions of multiple base models are used as input features to train a meta-learner
XGBoost	An optimized distributed gradient boosting[6] library designed for speed and performance, widely used in structured data machine learning tasks

7 APPENDICES

Figure 17:similarity

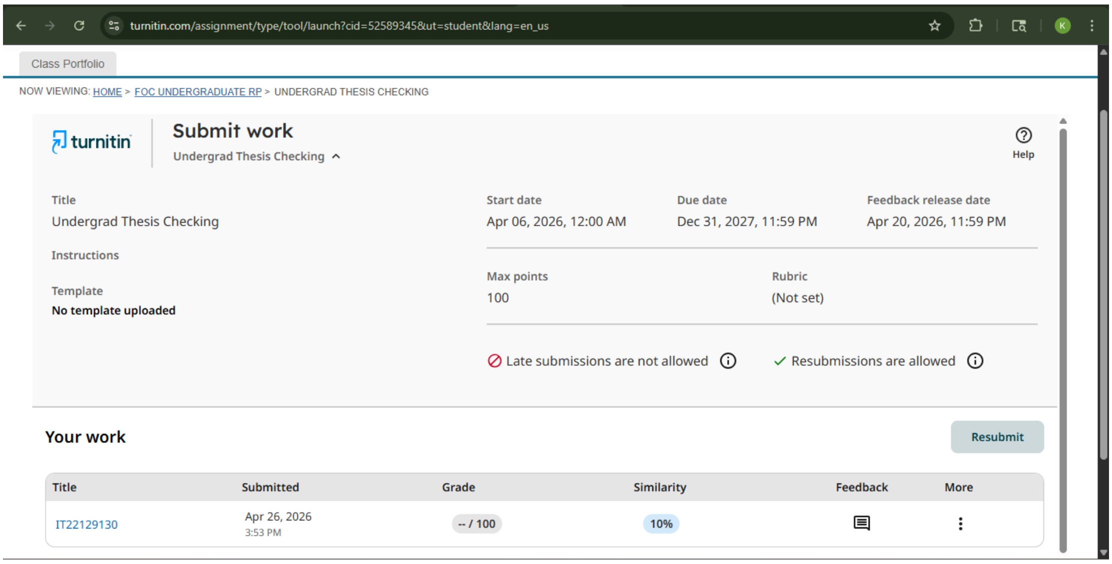


Figure 18:similarity

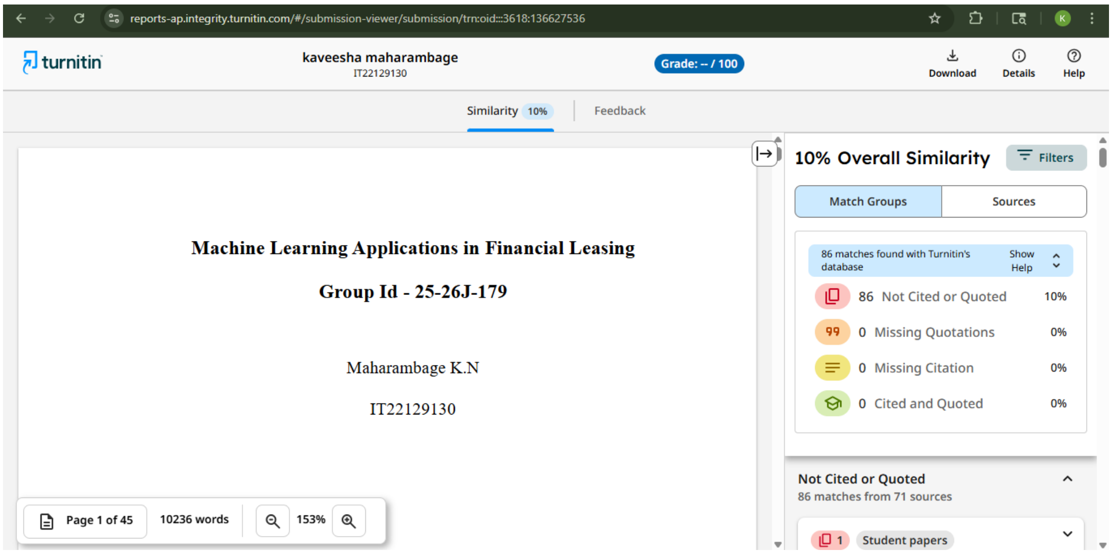
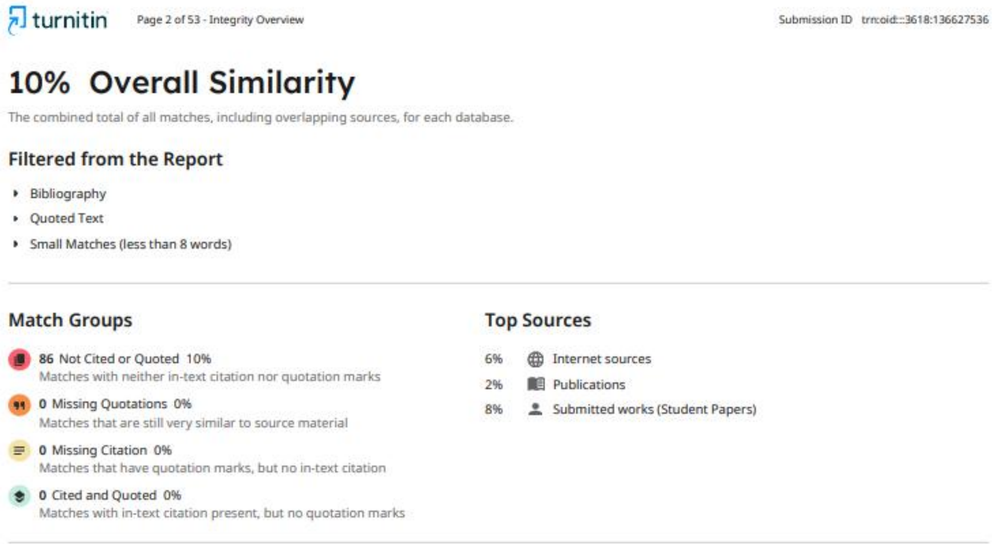


Figure 19:similarity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